Proofs About Sets & Quantification

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CS 0220 2022

February 11, 2022

Overview

Proving set equalities: set-element method

Theorem: For any sets A and B of elements in universe U, $\overline{A \cap B} = \overline{A} \cup \overline{B}$.

DeMorgan's Law again! Now, connects intersection and union instead of $\wedge$ and $\vee$.

An object $x \in \overline{A \cap B}$ if it is *not* in both A and B. Such an element must either not be in A or not be in B. It follows that such an element must be in $\overline{A} \cup \overline{B}$. Thus, we have $\overline{A \cap B} \subseteq \overline{A} \cup \overline{B}$.
Note also that an object $x \in \overline{A} \cup \overline{B}$ if it is either not in A or it is not in B. Such an element can't be in both A and B, therefore. Said another way, it must be in $\overline{A \cap B}$. Thus, we have $\overline{A} \cup \overline{B} \subseteq \overline{A \cap B}$.
Since both $\overline{A \cap B} \subseteq \overline{A} \cup \overline{B}$ and $\overline{A} \cup \overline{B} \subseteq \overline{A \cap B}$ are true, we know $\overline{A \cap B} = \overline{A} \cup \overline{B}$.

Fact about groups of people

Any two people have either met or not.

Given a group of people G, if all pairs of people in G have met, we'll call it a *club*. If no two people in G have met, we'll call them *strangers*.

Theorem. Every collection of 6 people includes a club of 3 people or a group of 3 strangers.

Does that seem true? Try some examples on the board.

Proof (Part 1)

The proof is by case analysis. Let x denote one of the six people. Let $R = G - \{x\}$ be the rest. There are two cases:

1. Among R, at least 3 have met x.
2. Among R, at least 3 have *not* met x.

At least one of these cases must hold. Since $|R|$ is odd, either more than half in R know x or less than half in R know x (and therefore more than half do not know x).

Case 1: At least 3 have met x. Let $J \subseteq R$ be those individuals. Two subcases:

1.1 No pair in J have met each other. So, J is a group of at least 3 strangers and the theorem holds in this subcase.

1.2 Some pair in J have met each other. That pair and x are a club of 3 people and the theorem holds in this subcase, too.

That covers Case 1!

Proof (Part 2)

Case 2: At least 3 have not met x. Let $J \subseteq R$ be those individuals. Two subcases:

2.1 Every pair in J have met each other. So, R is a club of at least size 3 and the theorem holds in this subcase.

2.2 Some pair in J haven't met each other. That pair and x are a group of strangers of 3 people and the theorem holds in this subcase, too.

That covers Case 2! It's kind of the inverse-video version of Case 1.

Since we showed that only these two cases can occur and the theorem holds in both, the theorem *always* holds.

Quantifiers, Revisited

Always True (universal quantification)
$\forall x \in \mathbb{R}, x^2 + 1 \geq 0.$

- For all $x \in D$, $P(x)$ is true.
- $P(x)$ is true for every x in the set D.

Sometimes True (existential quantification)
$\exists x \in \mathbb{Z}$, x is even and x is prime.

- There is an $x \in D$ such that $P(x)$ is true.
- $P(x)$ is true for some x in the set D.
- $P(x)$ is true for at least one $x \in D$.

Mixing quantifiers

Theorem (sparse squares): There's a perfect square arbitrarily far from its closest perfect square.

Clear? Maybe a tad vague. True? How say in math?

$$\forall d \in \mathbb{N}, \exists i \in \mathbb{N}, \forall j \in \mathbb{N},$$
$$(i \text{ is a perfect square}) \wedge (|i - j| \leq d \rightarrow \neg(j \text{ is a perfect square})).$$

The expressions nest inside each other. The order matters.

You can think of it like a little game. I'm claiming that you can pick any d you want. I'll then pick an i that's a perfect square AND no matter what j you pick that is within d values of i, j won't be a perfect square.

So, what's my winning strategy?

Any ambiguity is too many

"If you can identify any bird, you've got a talent."

1. If $\exists b$, you can identify b, then you've got a talent.
2. If $\forall b$, you can identify b, then you've got a talent.

"...statistics show that, in the UK, some brews a cup of tea every second. That person's name is Nigel."

1. $\forall t, \exists p$, p brews a cup of tea at second t
2. $\exists p, \forall t$, p brews a cup of tea at second t

More examples

"The whole article is not available."

- not $\forall$ article part x, x is available
- $\forall$ article part x, x is not available

Data privacy: consider an "anonymous" survey with a number of questions. How to pool the responses?

- For each response r, for each answer a on r, there is a person p who gave answer a.
- For each response r, there is a person p such that for each answer a on r, p gave answer a.

DeMorgan returns: Negating quantifiers

These two statements are equivalent:

- Not everyone likes coffee.
- There's someone who doesn't like coffee.

$\neg\forall x, P(x)$ is equivalent to $\exists x, \neg P(x)$.

Assertion about predicates

The formula$\exists x, \forall y, P(x, y)$ implies the formula $\forall y, \exists x, P(x, y)$.

If $\exists x, \forall y, P(x, y)$, there must be some specific x^* such that $\forall y, P(x^*, y)$. As a result, $\forall y, \exists x, P(x, y)$ because we can always choose x^* to be the selected x.

On the other hand,
$\forall y, \exists x, P(x, y)$ does not imply $\exists x, \forall y, P(x, y)$.

Let's come up with a *counterexample*: a property P where the first formula is true and the second is false.

We can describe P by a table saying for which values of x and y $P(x, y)$ is true:

	y_1	y_2	y_3
x_1	T	F	F
x_2	T	T	T
x_3	F	T	F

	y_1	y_2	y_3
x_1	T	F	F
x_2	T	F	T
x_3	F	T	F